

PROGRAM OF LINEAR SEARCH

```
#include<stdio.h>

# define size 20

int linear_search(int arr[] , int n , int target)
{
    for(int i = 0 ; i < n ; i++)
    {
        if(arr[i]== target)
        {
            return i;
        }
    }
    return -1;
}

int main()
{
    int arr[size] , target , n ;

    printf("\n enter the number of element in array : ");
    scanf("%d",&n);

    printf("\n enter the element in the array..");
    for(int i = 0 ; i < n; i++)
    {
        printf("\n enter the arr[%d] element : ",i+1);
        scanf("%d",&arr[i]);
    }

    printf("enter the target element : ");
    scanf("%d",&target);

    int pos = linear_search(arr , n ,target);

    if(pos == -1)
    {
        printf("\n the element is not found ");
    }
    else{
```

```
        printf("\n the element is found at %d position ",pos+1);  
    }  
}
```

OUTPUT :

enter the number of element in array : 2

enter the arr[1] element : 3

enter the arr[2] element : 2

enter the target element : 2

the element is found at 2 position

ALGORITHM :

LINEAR_SEARCH(A, N, VAL)

Step 1: [INITIALIZE] SET POS = -1

Step 2: [INITIALIZE] SET I = 1

Step 3: Repeat Step 4 while(I<=N)

Step 4: IF A[I] = VAL
 SET POS = I
 PRINT POS
 Go to Step 6
 [END OF IF]
 SET I = I + 1

 [END OF LOOP]

Step 5: IF POS = -1

 PRINT VALUE IS NOT PRESENT IN THE ARRAY

 [END OF IF]

Step 6: EXIT